BTCS301T — COMPUTER VISION

UNIT 1 — 20 QUESTIONS

Image Formation & Image Processing

Q1. In the pinhole camera model, an image is formed by light rays passing through:
- A. A single small aperture
- B. A convex lens array
- C. Two apertures
- D. A curved mirror

Q2. Radiometry, as used in image formation, primarily deals with the measurement of:
- A. Camera resolution
- B. Light energy/intensity
- C. Pixel coordinates
- D. Lens curvature

Q3. Digitization of an analog image consists of which two processes?
- A. Filtering and convolution
- B. Sampling and quantization
- C. Rotation and translation
- D. Compression and encoding

Q4. Spatial resolution of a digital image is primarily determined by:
- A. Quantization levels
- B. Sampling rate
- C. Colour depth
- D. Histogram spread

Q5. The number of gray levels used to represent pixel intensity is determined by:
- A. Sampling
- B. Quantization
- C. Convolution
- D. Radiometry

Q6. Which of the following transformations includes only rotation and translation (no scaling or shearing)?
- A. Affine transformation
- B. Projective transformation
- C. Rigid transformation
- D. Perspective transformation

Q7. A key property preserved by an affine transformation is:
- A. Angles between lines
- B. Parallelism of lines
- C. Absolute distances
- D. Pixel intensity values

Q8. Histogram equalization is primarily used to:
- A. Remove periodic noise
- B. Improve image contrast
- C. Detect edges

D. Estimate camera pose

Q9. Which of the following kernels is typically used for image blurring?

A. Sobel kernel
B. Laplacian kernel
C. Averaging/Gaussian kernel
D. Canny kernel

Q10. A high-pass spatial filter primarily enhances:

A. Smooth regions
B. Edges and fine details
C. Overall brightness
D. Colour saturation

Q11. The Fourier transform converts an image from the:

A. Frequency domain to spatial domain
B. Spatial domain to frequency domain
C. RGB domain to HSV domain
D. 2D domain to 1D domain

Q12. In the frequency domain, sharp edges in an image correspond to:

A. Low-frequency components
B. High-frequency components
C. Zero-frequency (DC) component only
D. No frequency component

Q13. The HSV colour transform separates an image into:

A. Red, Green, Blue
B. Hue, Saturation, Value
C. Luminance, Chrominance-1, Chrominance-2
D. Cyan, Magenta, Yellow

Q14. The image negative transform is given by $s = L - 1 - r$, where L is:

A. The number of gray levels
B. The image width
C. The kernel size
D. The focal length

Q15. Convolution in the spatial domain is equivalent to which operation in the frequency domain?

A. Addition
B. Subtraction
C. Multiplication
D. Division

Q16. Which pixel transform is commonly used for removing salt-and-pepper noise?

A. Median filter
B. Sobel filter
C. Fourier filter
D. Histogram equalization

Q17. An identity convolution kernel applied to an image will:

A. Blur the image
B. Sharpen the image
C. Leave the image unchanged

D. Invert the image

Q18. In the thin-lens model, the relationship between object distance, image distance, and focal length is derived from:

A. Similar triangles / lens equation
B. Bayes' theorem
C. Eigenvalue decomposition
D. Fourier series expansion

Q19. A projective transformation, unlike an affine transformation, does NOT necessarily preserve:

A. Straight lines
B. Parallelism of lines
C. Connectivity
D. Pixel colour

Q20. Removing high-frequency periodic noise from an image is most effectively done using:

A. Spatial-domain averaging only
B. Frequency-domain (Fourier) filtering
C. Histogram stretching
D. Morphological dilation